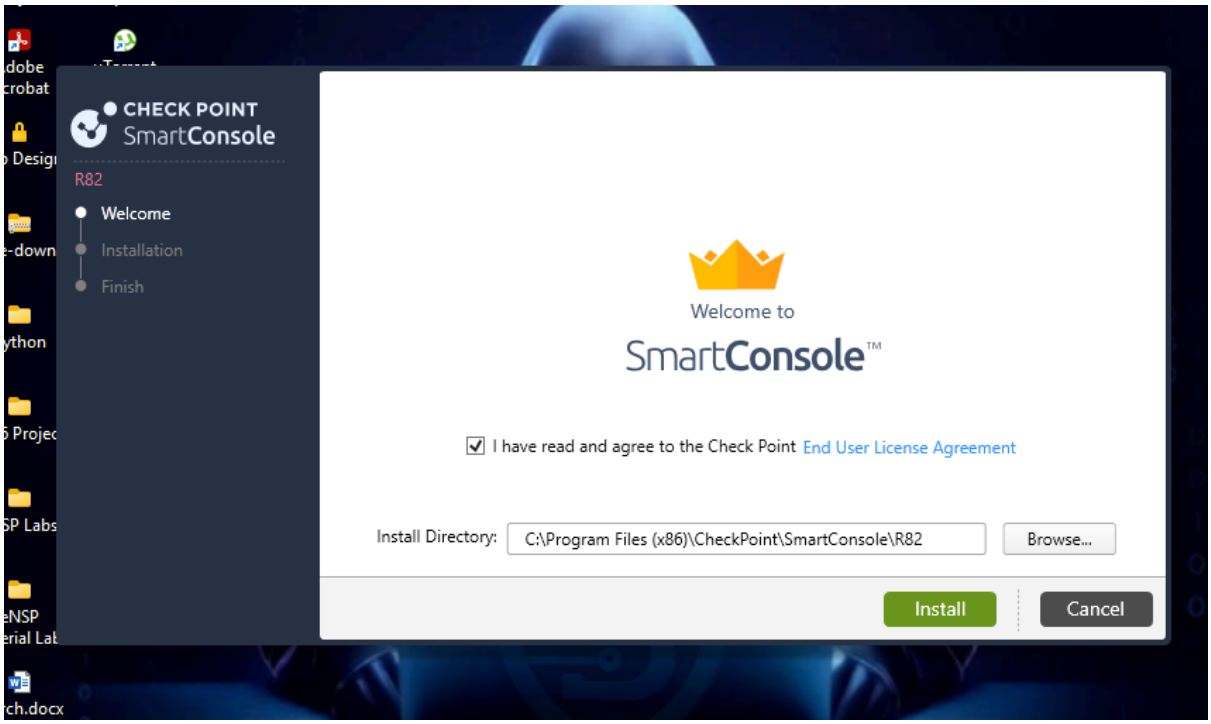
FULL NAME	**MUSA SHUAIBU**
REGISTRATION NUMBER	**C11/26/CCSA/17045**
SMARTCONSOLE AND A-SMS SERVER	**ICDFA provided**
VM HOSTNAME	**A-SMS, SmartConsule and Geia Portal**

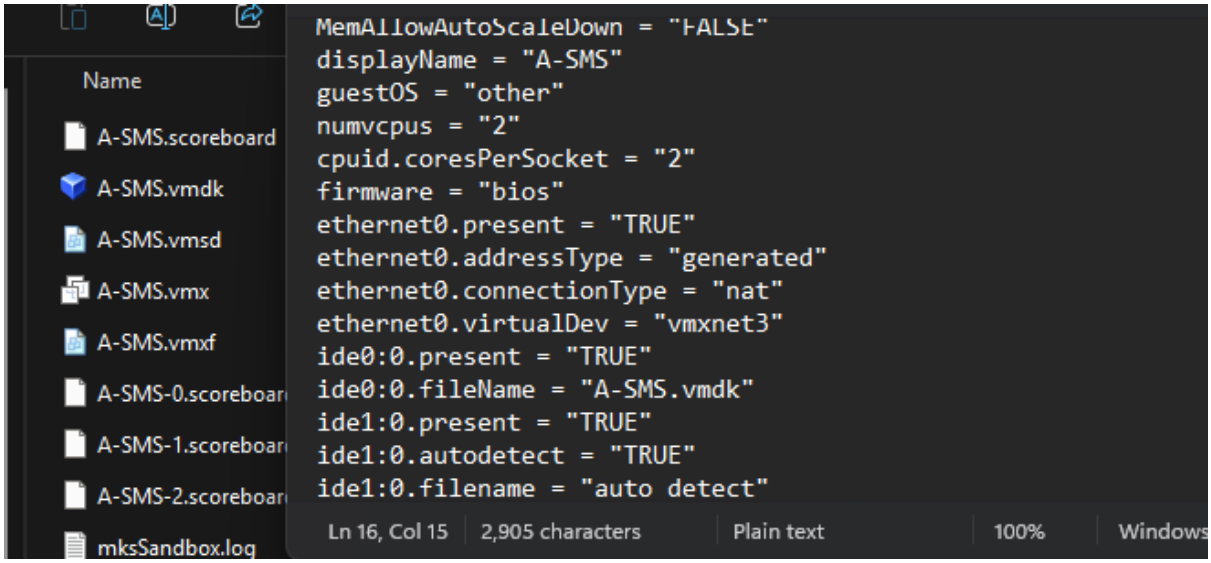
PART A PREPARE VMWARE AND THE PRIVATE NETWORK

Step 1: Install the Windows tools



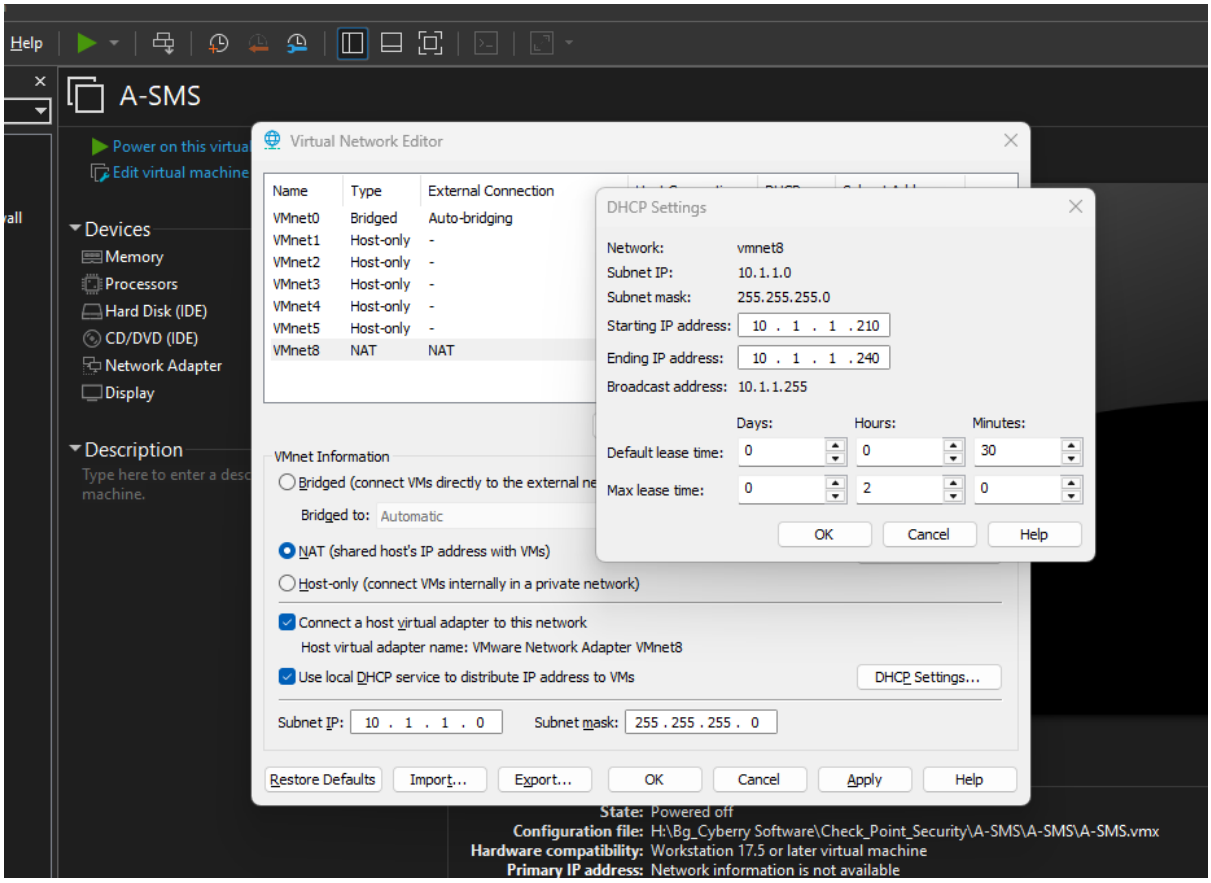


Step 2: Load A SMS and check its adapter

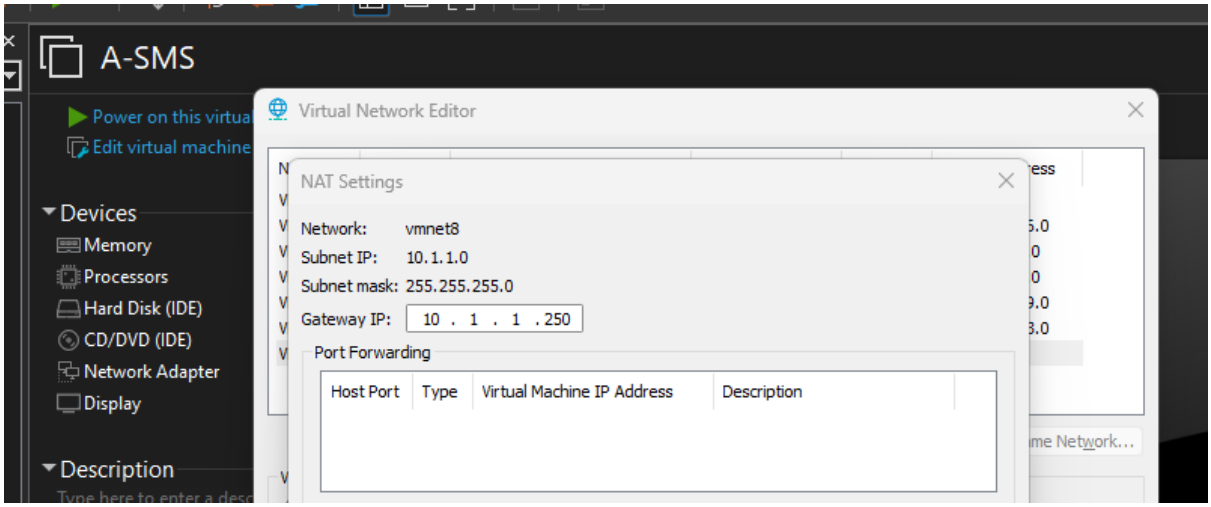


BRIEF EXPLANATION: This Confirm that the Extract A-SMS has the command **ethernet0.virtualDev = "vmxnet3"** added and was successfully saved

Step 3: Set the VMnet8 network



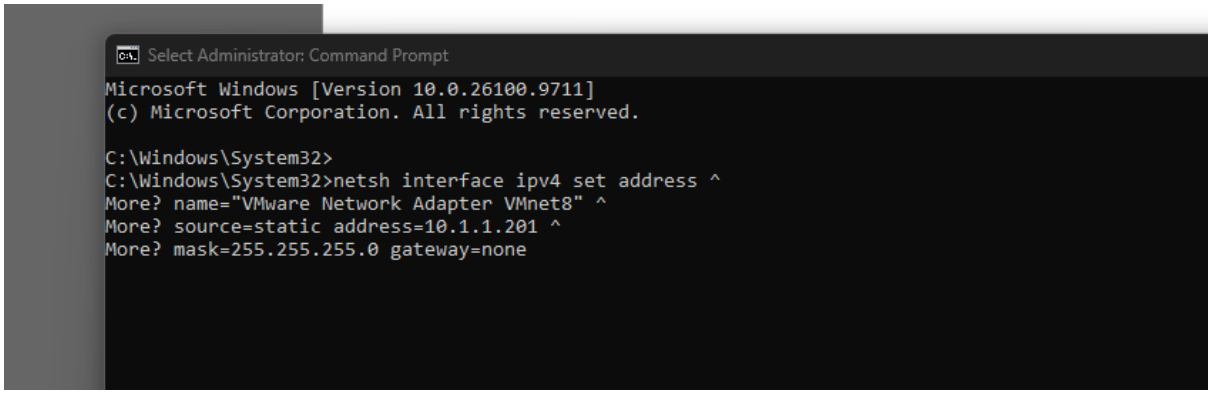
BRIEF EXPLANATION: This shows that the ip address of VMnet8 is set to a subnet of 10.1.1.0/24 also the DHCP server is been set to allocate ip address to the to host, with same subnet ranging from 10.1.1.210 to 10.1.1.240

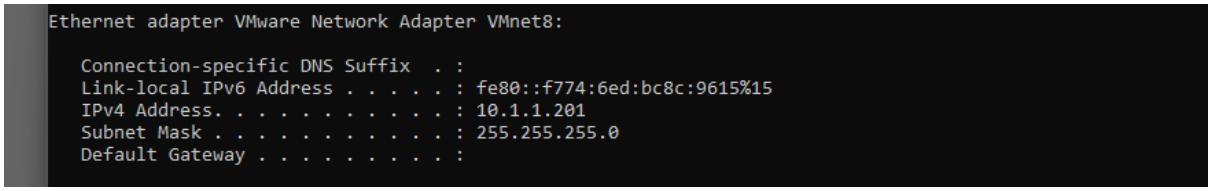


BRIEF EXPLANATION: This shows that the NAT is been set with a gateway of the same subnet with the VMnet8 so that traffic can pass to the internet

PART A SET THE ADDRESSES AND VERIFY GAIA

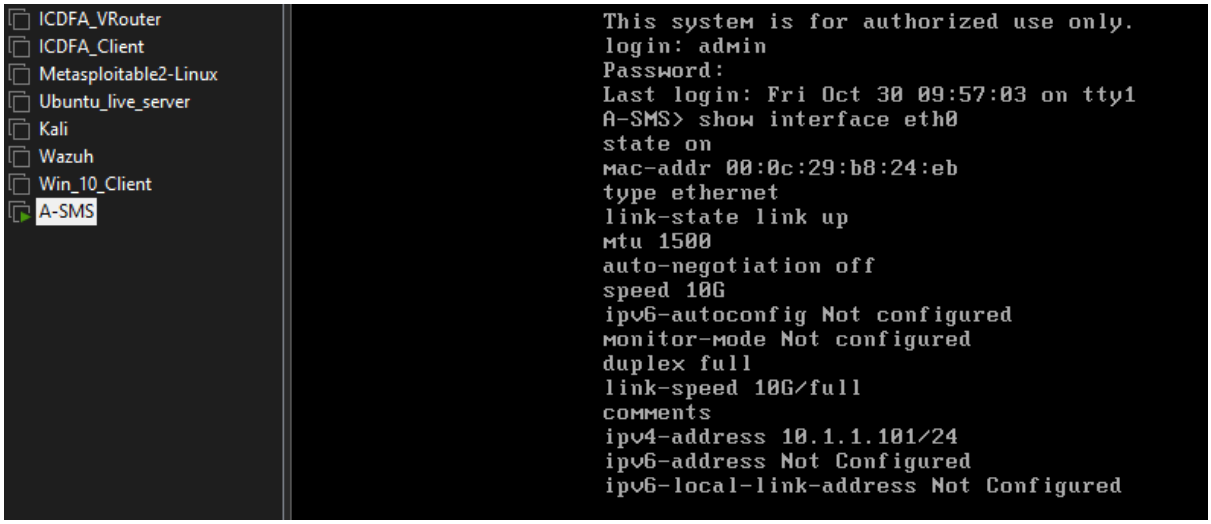
Step 4: Set the Windows VMnet8 address





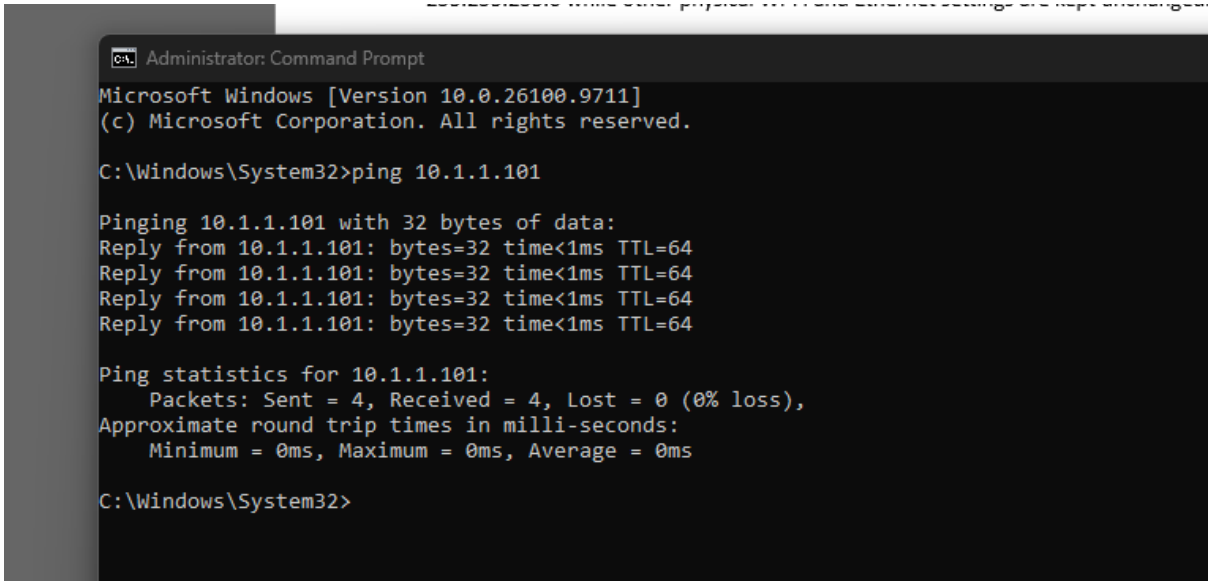
BRIEF EXPLANATION: This shows and Confirm that the VMnet8 shows 10.1.1.201 and mask 255.255.255.0 while other physical Wi-Fi and Ethernet settings are kept unchanged.

Step 5 Boot A SMS and inspect eth0

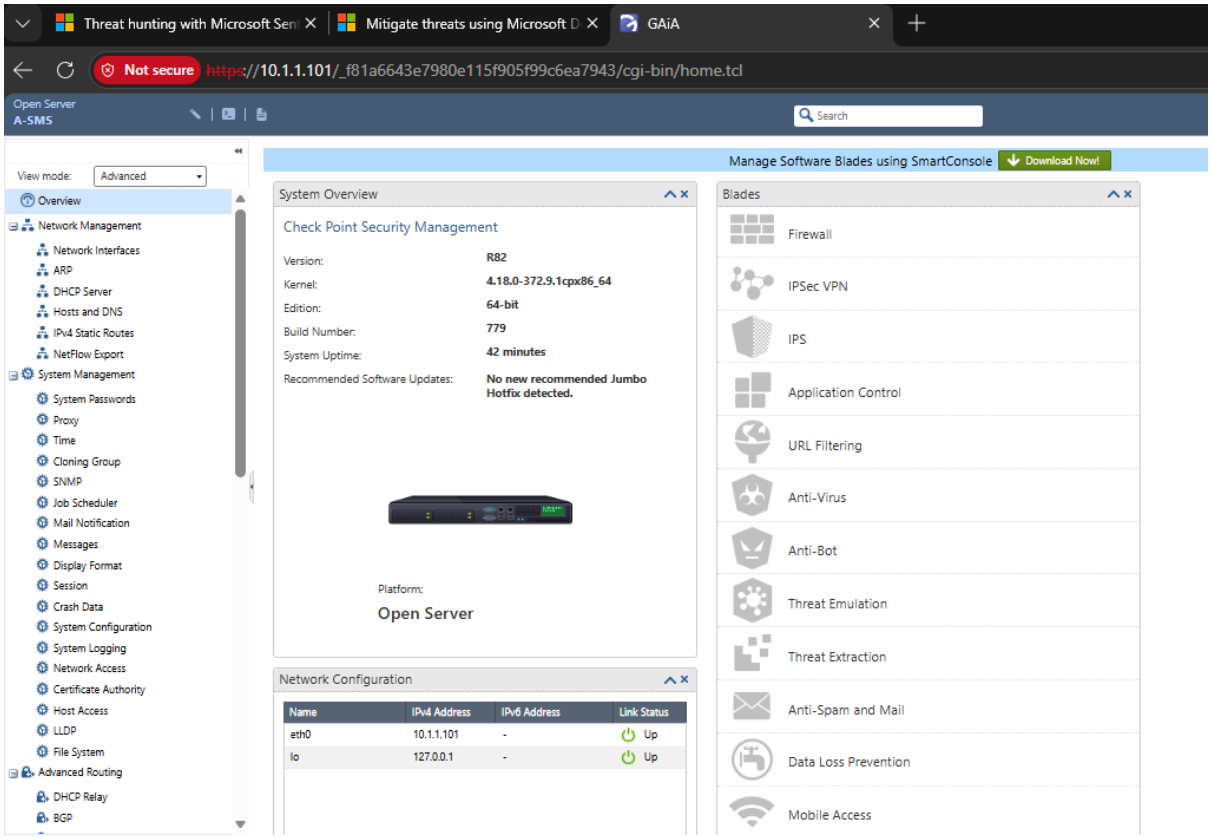


BRIEF EXPLANATION: This shows that the interface eth0 ipv4-address was already configured with ip address 10.1.1.101 and mask-length 24

Step 6 Check access from Windows



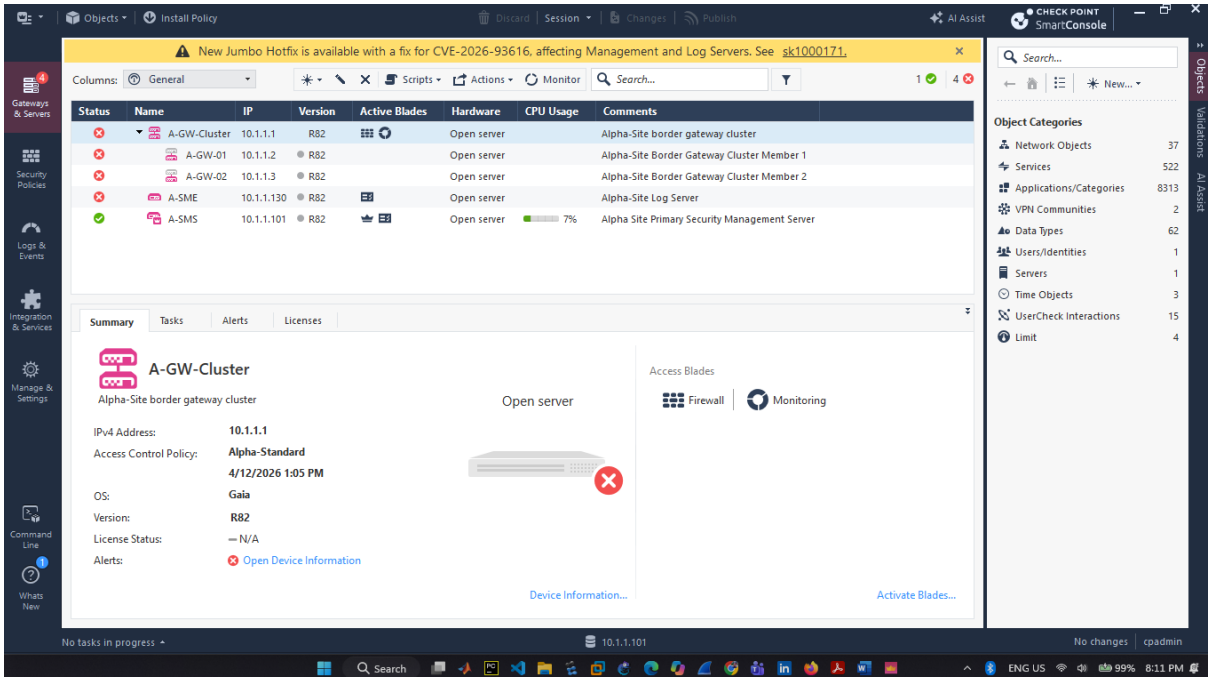
BRIEF EXPLANATION: This shows that the windows has successfully ping the security management sever with ip address of 10.1.1.101



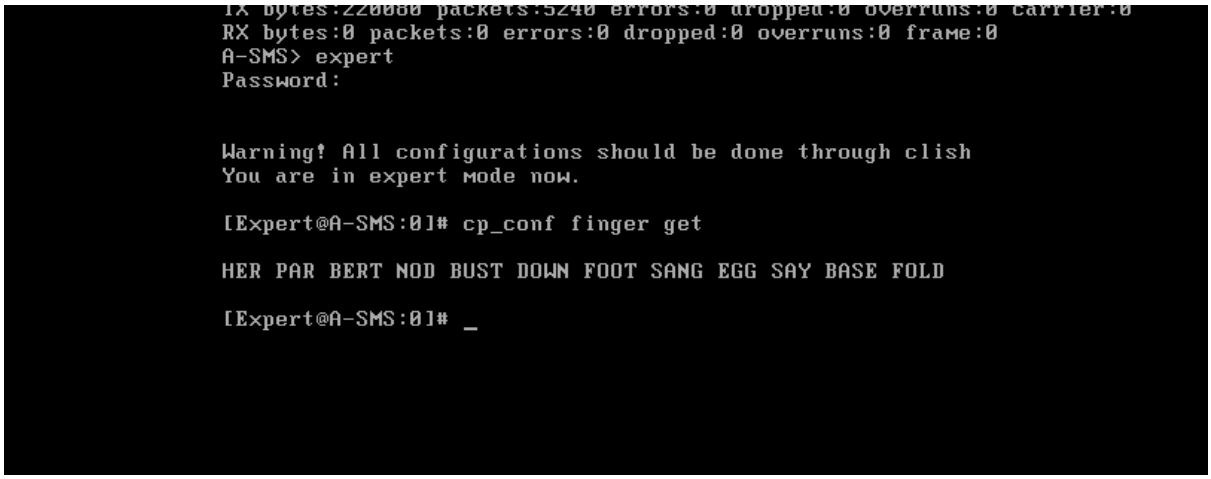
BRIEF EXPLANATION: This shows that the Gaia login page was successfully accessible through the windows web browser using the URL https://10.1.1.101 which is the ip address of the security management sever

PART A CONNECT SMARTCONSOLE

Step 7: Verify the server and sign in



BRIEF EXPLANATION: This shows that the SmartConsole was launched and connected to the Security Management Server on ip address of 10.1.1.101 and other gateways appear unavailable because only the A-SMS sever is power on



BRIEF EXPLANATION: This shows that the command **cp_conf finger get** is used to display the ICA fingerprint, the fingerprint was successfully retrieved as

HER PAR BERT NOB BUST DOWN FOOT SANG EGG SAY BASE FOLD

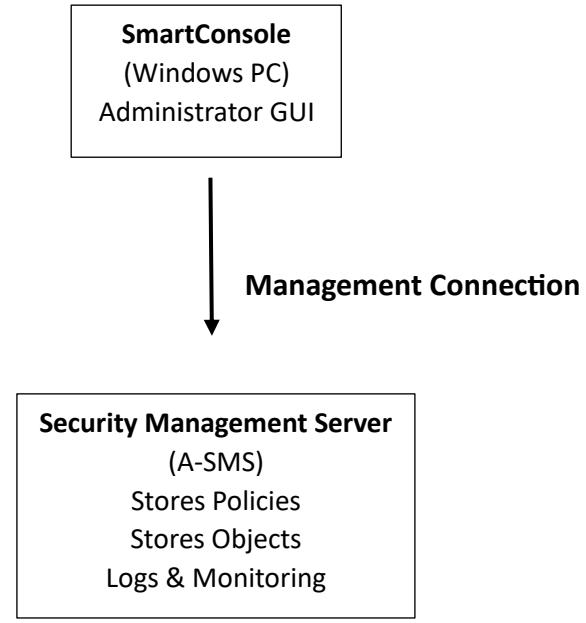
And this fingerprint can be used to verify the identity of the Security Management Server before establishing trust with SmartConsole

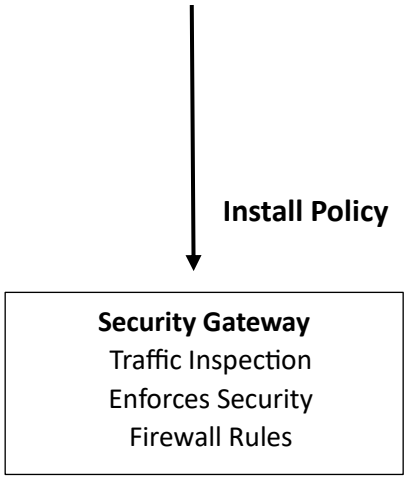
PART B EXPLAIN THE CHECK POINT ARCHITECTURE

B1 EXPLAIN THE ARCHITECTURE

Draw a labelled three-tier diagram showing SmartConsole on Windows, the Security Management Server and a Security Gateway. Explain each role, where policy and object definitions are maintained, and the difference between changing a policy and enforcing it.

Answer





SmartConsole is the graphical administration tool installed on a Windows computer. Security administrators use it to create security policies, manage network objects, monitor events, review logs, and configure Check Point security settings.

Security Management Server is the central management component. In our lab, this is the **A-SMS server (10.1.1.101)**. It stores:

Security policies
Network objects
User definitions
Logs and events
Administrator configurations

All policy changes are created and saved on the Management Server before being deployed.

Security Gateway is the enforcement point. It inspects network traffic and applies the security rules defined by the Management Server. The gateway determines whether traffic is allowed, blocked, or logged.

B2 EXPLAIN MANAGEMENT SERVICES AND DEPLOYMENT

Explain three Security Management Server functions. Compare an on-premises management deployment with Smart-1 Cloud, then recommend one for an organisation that wants less on-premises hardware maintenance. State one tradeoff.

Marks: three explained functions 3; comparison and justified recommendation 2.

ANSWER

1) Security Management Server (SMS) is the central component responsible for managing the Check Point environment.

Its functions include:

- Storing security policies
- Managing network objects
- Managing administrators and permissions
- Collecting and storing logs
- Deploying policies to Security Gateways

- Monitoring security events

2) SmartConsole is a Windows-based application used by administrators to manage Check Point security systems.

Using SmartConsole, an administrator can:

- Create firewall rules
- Configure VPNs
- Define network objects
- Monitor logs
- Install policies on gateways
- Manage users and administrators

SmartConsole connects to the Security Management Server, not directly to the Security Gateway.

3) Security Gateway is the device that enforces security policies.

Responsibilities include:

- Inspecting network traffic
- Allowing authorized traffic
- Blocking unauthorized traffic
- Generating security logs
- Detecting threats

The gateway receives policies from the Management Server and applies them to traffic passing through it.

On-Premises Management Deployment

- The Security Management Server is installed and managed within the organization's own data center or server room.
- The organization is responsible for hardware, software updates, backups, maintenance, and troubleshooting.
- Provides greater control over the management infrastructure and data.

Smart-1 Cloud

- Smart-1 Cloud is a cloud-based management platform provided by Check Point.
- Check Point manages the infrastructure, software updates, maintenance, and availability.
- Administrators access management services through the cloud without maintaining local management hardware.

Recommendation For an organization that wants **less on-premises hardware maintenance**, I would recommend **Smart-1 Cloud**.

Reasons:

- No need to purchase or maintain management servers.
- Reduced administrative overhead.
- Automatic updates and maintenance by Check Point.
- Faster deployment and scalability.

One Tradeoff: A tradeoff of using **Smart-1 Cloud** is that the organization has **less direct control over the management infrastructure** and may depend on internet connectivity to access management services.

B3 CHOOSE THE CORRECT ADMINISTRATION INTERFACE

Choose and justify the tool for each task: inspect eth0 and its subnet mask; review an Access Control rule; inspect operating-system resource use; view security logs. Then distinguish Gaia Clish, Expert mode and SmartConsole's Command Line panel.

Marks: four appropriate tool choices with reasons 3; correct command-context distinction 2.

ANSWER

1. Inspect eth0 and its subnet mask

Tool: Gaia Clish

Reason: Gaia Clish is used for network and system administration tasks. Commands such as:

The command **show interface eth0** will display the interface IP address, subnet mask, administrative state, and link status. This makes Clish the appropriate tool for inspecting network interface configuration.

Example

Gaia Clish

show interface eth0

show version all

show uptime

Expert Mode

whoami

pwd

cp_conf finger get

2. Review an Access Control Rule

Tool: SmartConsole

Reason: Access Control policies are managed through SmartConsole. Administrators can open the Security Policies view and inspect rules including Source, Destination, Services, Action, and Track settings. Gaia Clish and Expert mode are not the primary interfaces for reviewing firewall policy rules.

SmartConsole Command Line Panel

Example

ip route

ping -c 4 10.1.1.101

3. Inspect Operating System Resource Use

Tool: Gaia Portal or Expert Mode

Reason: System resources such as CPU usage, memory utilization, uptime, and running processes are operating system information. Gaia Portal provides graphical monitoring, while Expert mode provides Linux-based diagnostic commands for deeper inspection.

4. View Security Logs

Tool: SmartConsole

Reason: Security logs are viewed through the **Logs & Events** section of SmartConsole. Administrators can review traffic activity, policy actions, security alerts, and audit events from the management environment.

B4 EXPLAIN SECURITY FEATURES AND POLICY HANDLING

Choose two Security Gateway Software Blades or features covered in the course. Give a realistic use case for each. Explain why publishing a management session and installing policy on a gateway are separate actions.

Marks: features and relevant use cases 2; accurate publishing and installation explanation 3.

ANSWER

The Two Security Gateway Software Blades/Features and Their Use Cases are as follows

1) Firewall Blade controls network traffic by enforcing security rules that determine which connections are allowed or blocked.

Use Case: A company allows employees to access web servers on the Internet using HTTP/HTTPS while blocking unauthorized services such as Telnet. The Firewall Blade inspects the traffic and applies the configured access rules.

2) IPS Blade detects and prevents known attacks and malicious activities by inspecting network traffic for attack signatures and suspicious behaviour.

Use Case: If an attacker attempts to exploit a known vulnerability in a web server, the IPS Blade can identify the attack pattern and block the traffic before it reaches the server

Why Publishing and Installing Policy Are Separate Actions

In Check Point, **publishing** and **installing policy** are two different processes.

Publishing a Session

When an administrator makes changes in SmartConsole, the changes are initially stored only in that administrator's private session.

Selecting **Publish**:

- Saves the changes to the Security Management Server.
- Makes the changes visible to other administrators.
- Does **not** send the policy to the Security Gateway.

Installing Policy:

- Takes the published policy from the Security Management Server.
- Transfers it to the Security Gateway.
- Causes the gateway to begin enforcing the new rules.

B5 Interpret your own lab evidence

Explain why the private VMnet8 addresses can remain fixed when Wi-Fi changes and why enabled VMware DHCP does not alter the static A-SMS address. Explain what ping proves compared with SmartConsole login. Finally, explain why an unavailable gateway object or empty log view needs further evidence before you diagnose a fault.

Marks: private networking and DHCP 2; connectivity evidence 1; reasoned interpretation of status and logs 2.

ANSWER

1. Why VMnet8 Addresses Remain Fixed When Wi-Fi Changes

VMware **VMnet8** is a virtual NAT network created on the host computer. The virtual machines communicate through this private virtual network, which is independent of the Wi-Fi network used by the host.

For example:

A-SMS = 10.1.1.101

A-GW-01 = 10.1.1.2

A-GW-02 = 10.1.1.3

These addresses remain the same even if the host computer connects to a different Wi-Fi network because they belong to the VMware virtual network rather than the physical network.

2. Why VMware DHCP Does Not Change the A-SMS Address

The A-SMS server is configured with a **static IP address (10.1.1.101)**.

DHCP only assigns addresses to devices configured to obtain an IP address automatically. Since A-SMS uses a manually configured static address, VMware DHCP will not overwrite or change it.

Therefore:

Static IP = Remains fixed

DHCP IP = Assigned automatically

3. What Ping Proves Compared with SmartConsole Login

Ping

A successful ping proves:

- IP connectivity exists.
- The destination device is reachable on the network.
- ICMP traffic is passing between devices.

Example:

ping 10.1.1.101

A successful reply shows that the Security Management Server can be reached.

SmartConsole Login

A successful SmartConsole login proves more than ping:

- Network connectivity exists.
- The Security Management Server application is running.
- Authentication is working.
- Management services are responding correctly.

Therefore, SmartConsole connectivity provides stronger evidence than a simple ping test.

4. Why an Unavailable Gateway or Empty Log View Needs Further Investigation

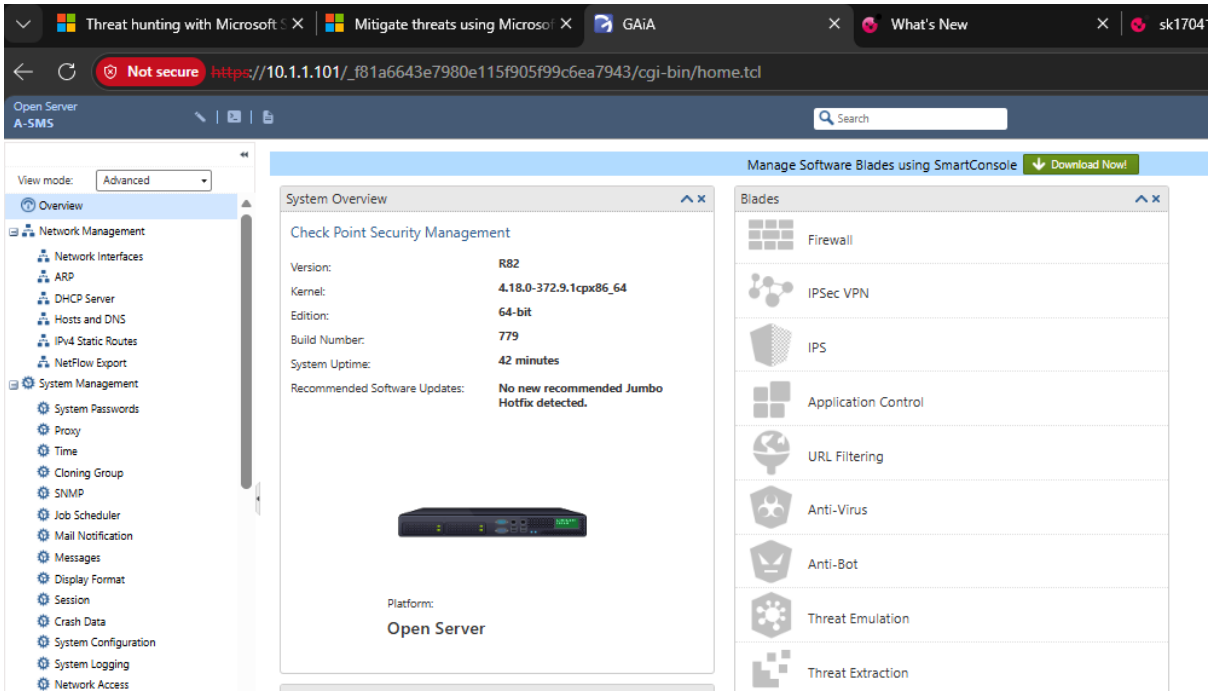
An unavailable gateway object or an empty log view does **not automatically mean a fault exists**.

Examples:

- The gateway VM may be powered off.
- The gateway may not yet be connected to management.
- No traffic may have occurred yet.
- No policy has been installed.
- Logging may not be enabled for the relevant rules.

PART C EXPLORE GAIA PORTAL

C1 INSPECT THE SYSTEM OVERVIEW

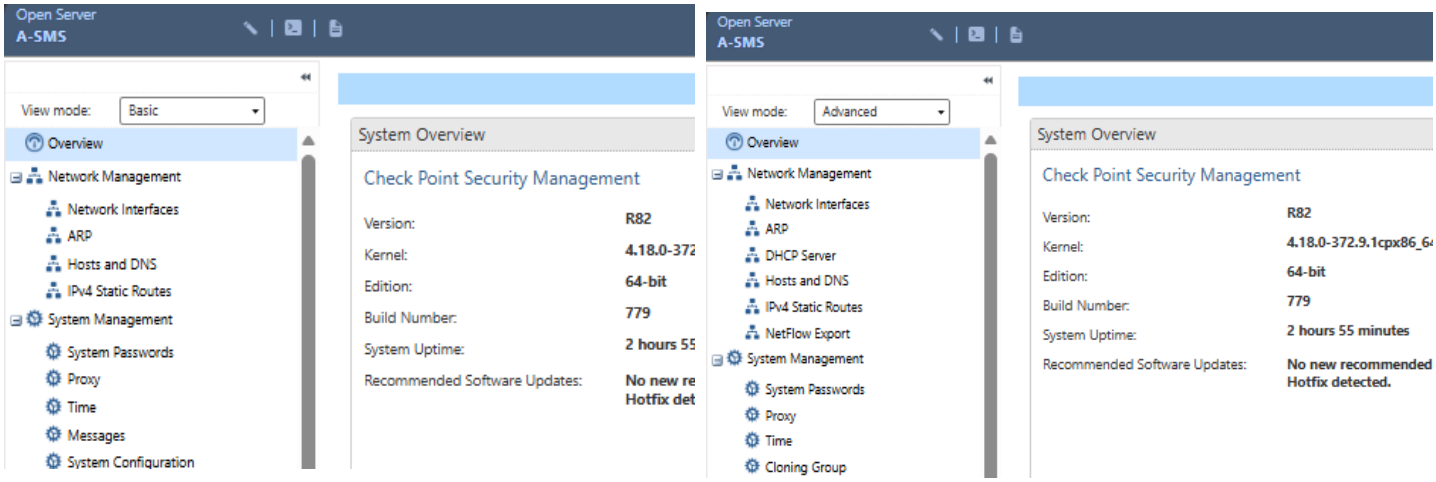


BRIEF EXPLANATION: This shows that the log in via https://10.1.1.101 with the following information

Hostname: A-SMS
Product Release: R82
Build Number: 779
Platform: Open Server
System Uptime: 42 minutes
Management IP Address: 10.1.1.101

Also, CPU and memory usage are dynamic values that change continuously based on system activity.

C2 COMPARE THE NAVIGATION MODES



Basic View vs Advanced View

Basic View

- Provides a simplified interface.
- Displays only the most commonly used configuration options.
- Easier for new administrators to navigate.
- Reduces clutter by hiding advanced features.

Advanced View

- Displays all available configuration and administration options.
- Provides access to detailed system, network, routing, security, and service settings.
- Intended for experienced administrators who need full control of the system.

Three Navigation Areas and Their Purposes

1. Overview

Purpose:

- Displays system status and summary information.
- Shows version, build number, uptime, software blades, and network status.

2. Network Management

Purpose:

- Configure and manage network settings.
- Includes interfaces, ARP, DHCP Server, Hosts and DNS, and IPv4 Static Routes.

3. System Management

Purpose:

- Manage system-level settings.

- Includes passwords, time settings, SNMP, logging, certificates, backup, and system configuration.

Option Available Only in Advanced View

Advanced Routing

– DHCP Relay

– BGP

appear in the navigation tree. These advanced routing features are typically not available in Basic View and are examples of options exposed in Advanced View.

C3 Inspect the management interface

Interface Details (eth0)

Item	Value
Interface Name	eth0
IPv4 Address	10.1.1.101
Subnet Mask	255.255.255.0
Administrative State	Up
Link Status	Up
MAC Address	Not shown in the current view

Comparison with Part A / Clish

The IPv4 address of eth0 is:

10.1.1.101

This matches the management server address used earlier in:

SmartConsole connection, Gaia Portal login and Clish interface checks

C4 Inspect routes and DNS

From Network Management IPv4 Static Route Information

IPv4 Static Routes

Item	Value
Destination	Default
Next Hop Type	Normal
Gateway	10.1.1.1
Rank	60

Ping Monitoring	No

Default Gateway: 10.1.1.1

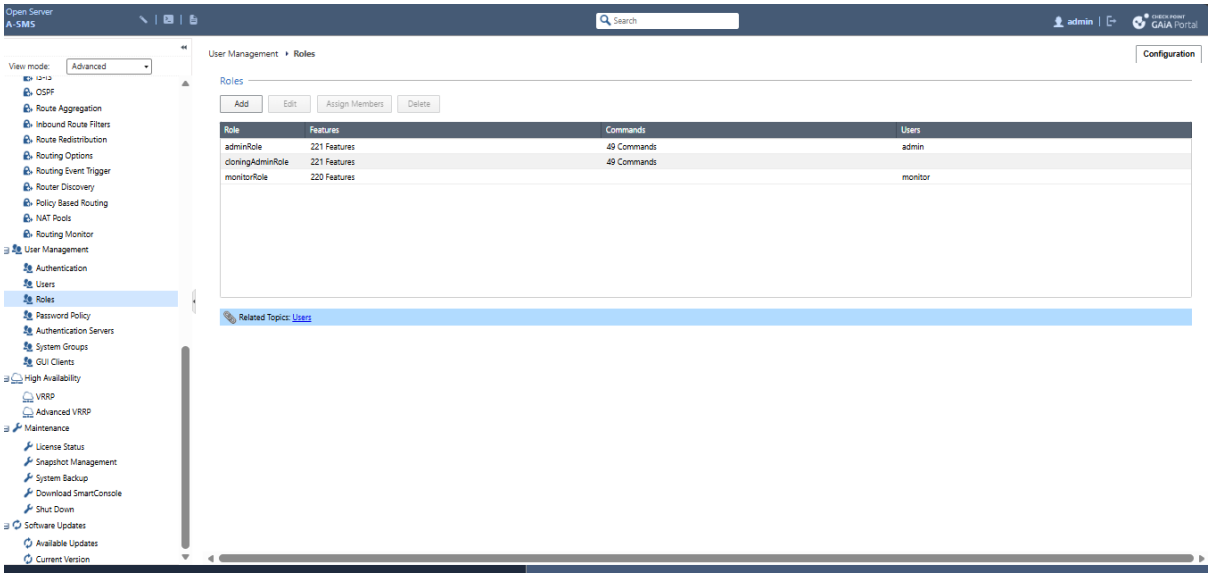
DNS Configuration

From **Network Management → Hosts and DNS**:

DNS Server	**IP Address**
Primary DNS Server	168.63.129.16
Secondary DNS Server	192.168.11.101
Tertiary DNS Server	1.1.1.1

Also, the default gateway and DNS servers were configured, however Internet connectivity was not directly tested during this task because Local access through Gaia Portal and SmartConsole does not prove Internet access.

C5 Locate maintenance and administration functions

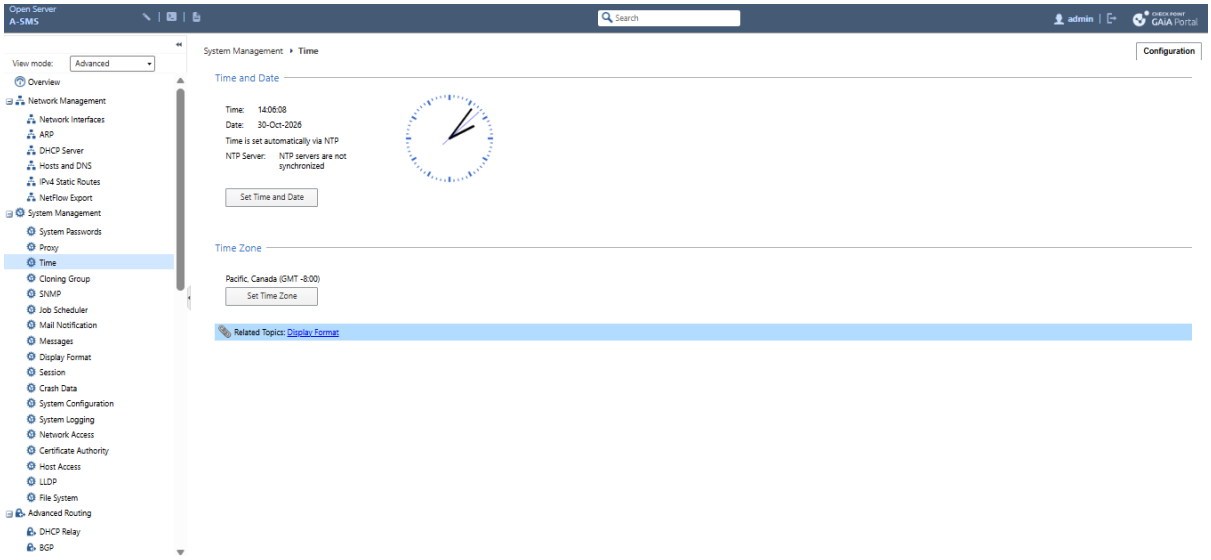


Purpose:

- Manage administrator roles and permissions.
- Assign privileges to users.
- Control administrative access levels.

Observed Roles:

- adminRole
- cloningAdminRole
- monitorRole



Purpose:

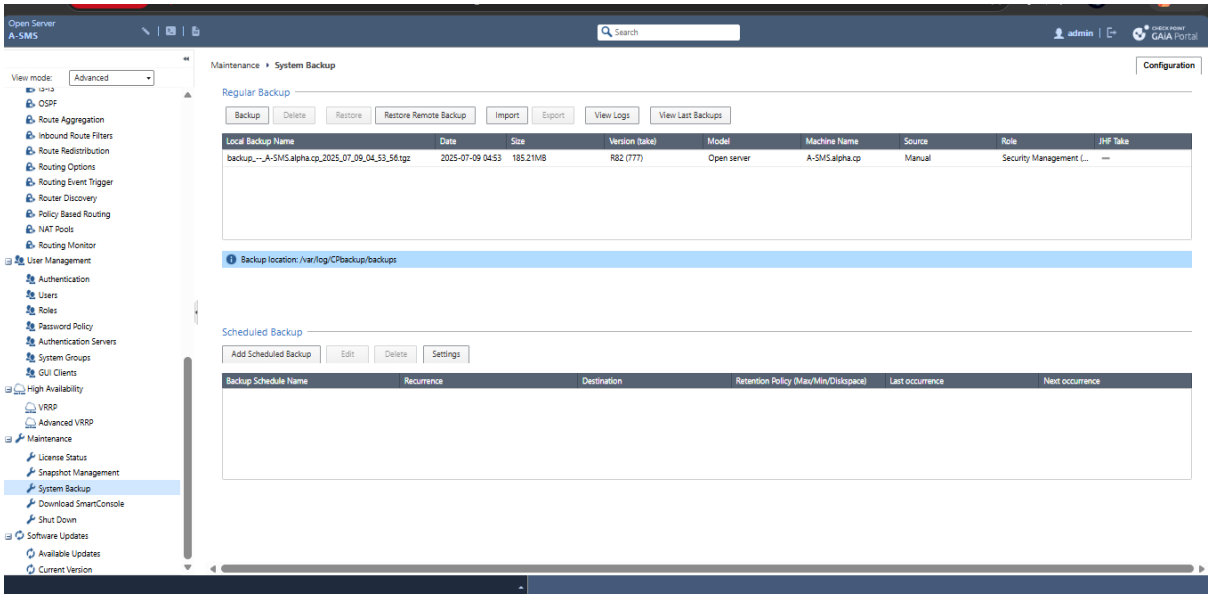
- Configure date and time.
- Configure NTP synchronization.
- Set timezone.

Observed:

Date: 30-Oct-2026

NTP Status: NTP servers are not synchronized

Time Zone: Pacific, Canada (GMT -8:00)



Purpose:

- Create configuration backups.
- Restore backups.
- Manage backup schedules.

Observed Backup:

Machine Name: A-SMS.alpha.cp

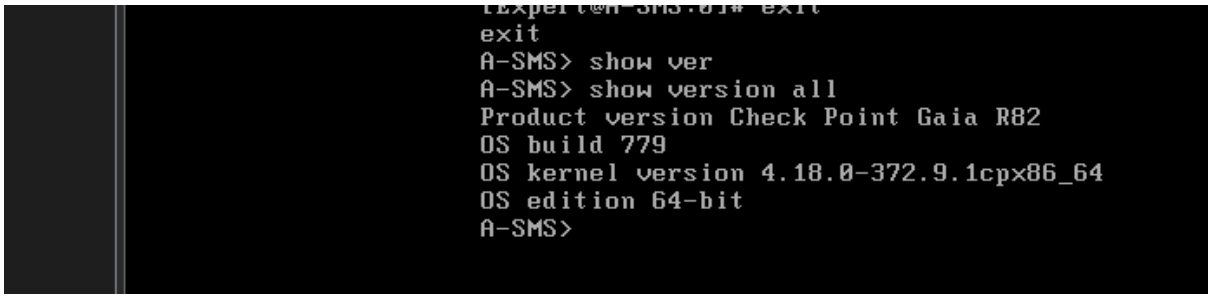
Version: R82 (777)

Size: 185.21 MB

Source: Manual

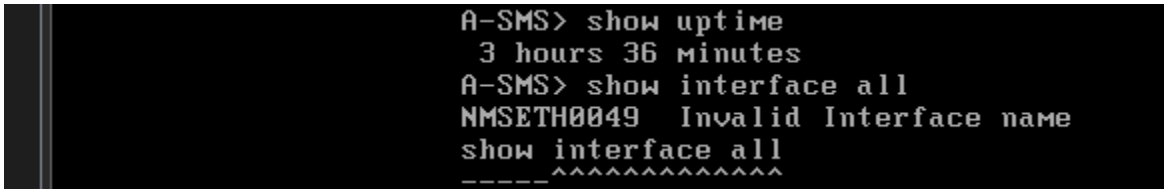
PART C INSPECT GAIA CLISH AND EXPERT MODE

C6 Collect system information in Clish

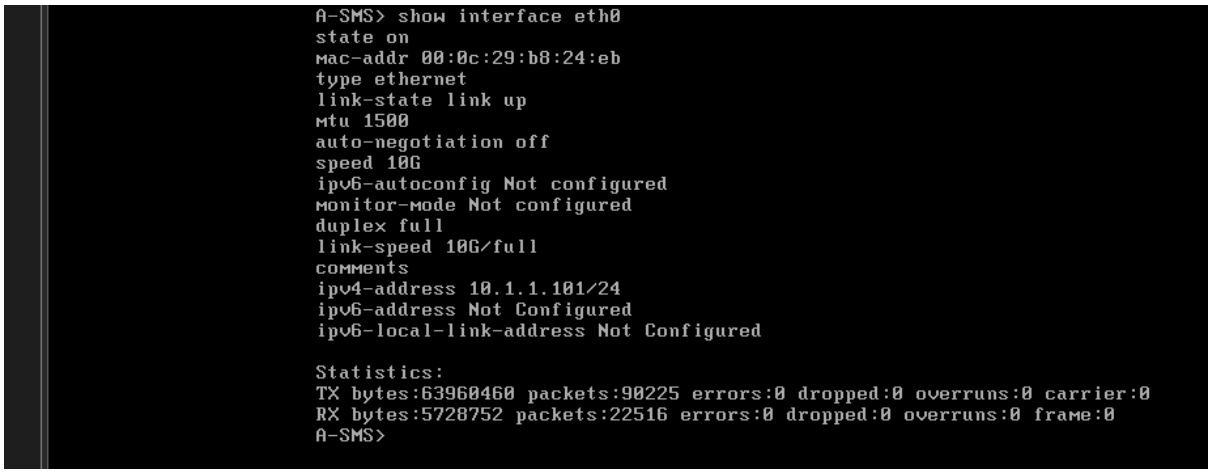


Brief Explanation: The reveals the following information

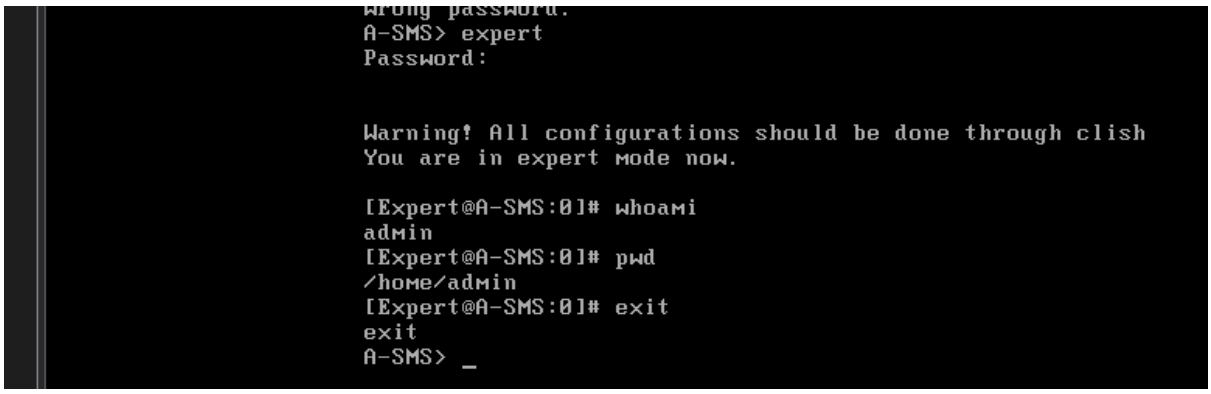
Product Release: **R82**
Build Number: **779**
Kernel Version: **4.18.0-372.9.1cpx86_64**
Operating System: **64-bit Gaia**



Brief Explanation: Indicates how long the A-SMS server has been running since the last reboot.



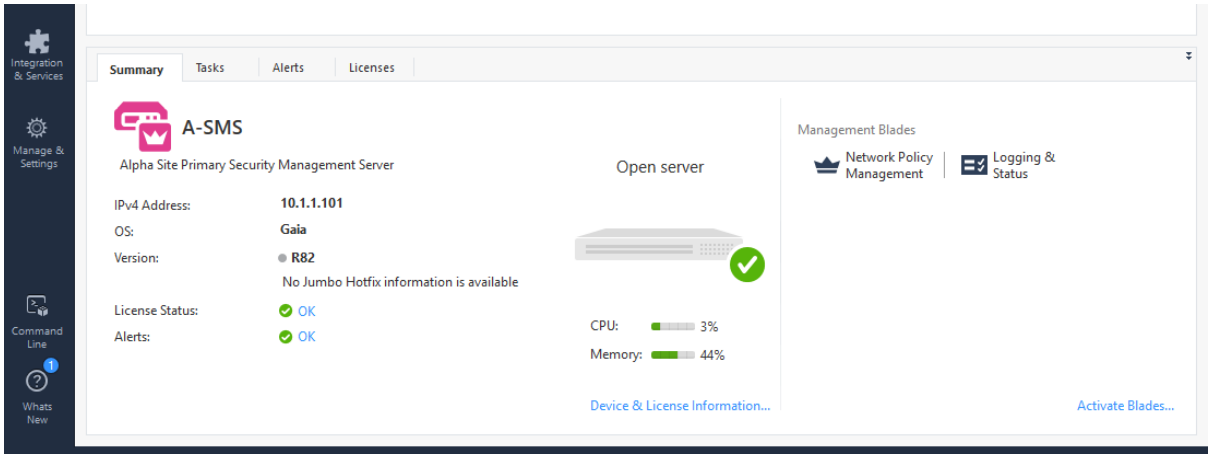
C7 Recognise the Expert shell

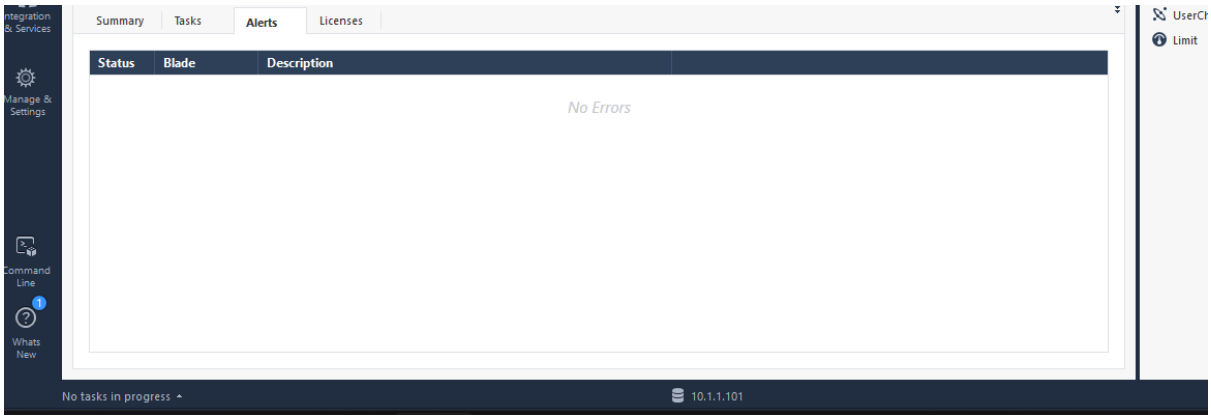


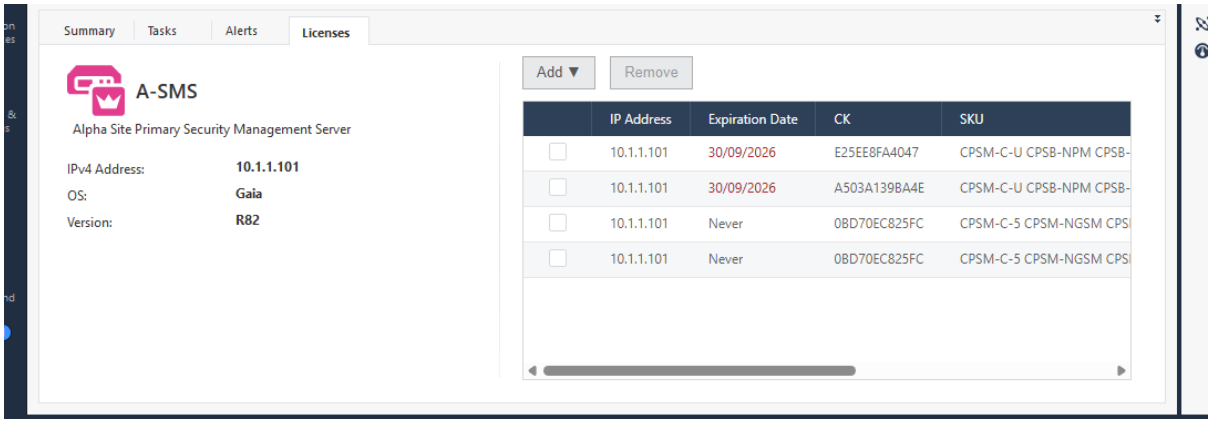
Expert mode was entered using the expert command. The prompt changed from A-SMS> in Clish to [Expert@A-SMS:0]#. The command whoami returned admin which confirming the active user account, and pwd returned /home/admin, showing the current working directory. The exit command returned the session to the Clish prompt.

PART C NAVIGATE SMARTCONSOLE

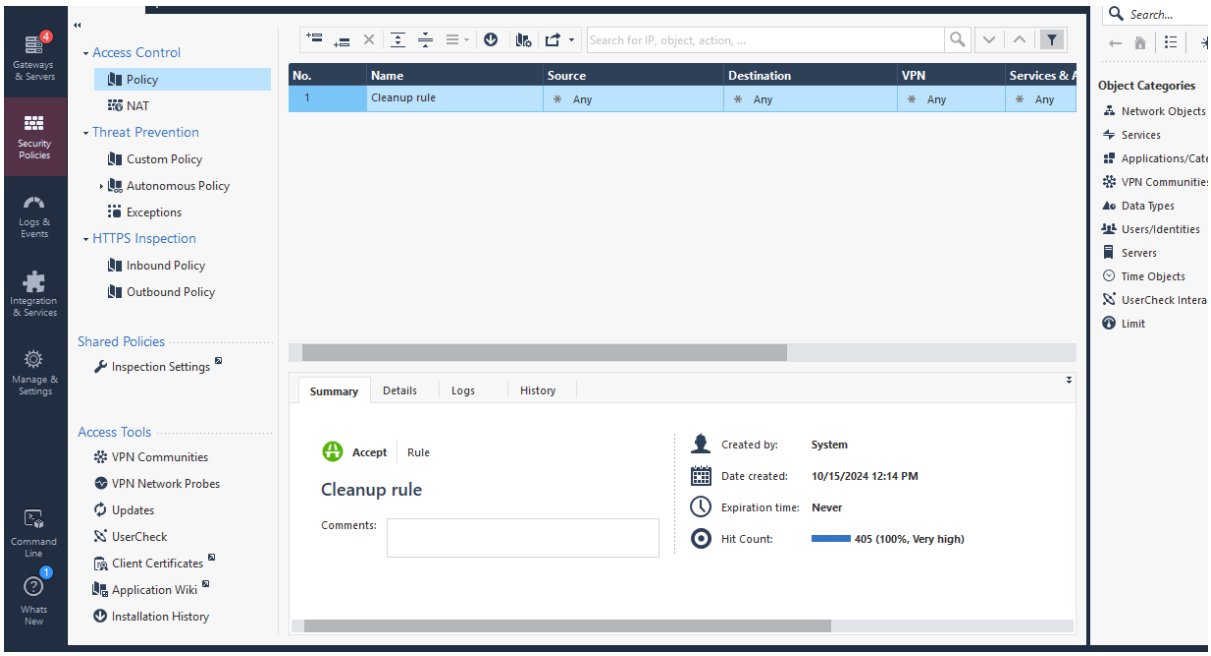
C9 Inspect Gateways and Servers







C10 Inspect an existing policy



C12 Inspect Logs and Events

